The Department of Mechanical Engineering is part of the Faculty of Engineering and Architecture and is fully accredited by the Pakistan Engineering Council. Its core technological focus involves providing engineering solutions for product and process development, ranging from micro-component design to massive industrial plants, machinery, and vehicles. Key areas of this focus include transportation, healthcare infrastructure, construction, thermal sensors or devices, and robotics. Graduates of the department possess highly diverse skills, securing career opportunities across various sectors such as manufacturing, automotive, power generation, construction, biomedical, aerospace, process industries, resource management, and oil and gas. The vision of the department is to be a highly reputed Mechanical Engineering faculty that offers its own unique blend of technical specializations and research opportunities. Its mission is to provide a modern, academically stimulating environment that produces socially responsible young professionals who can solve mechanical engineering challenges and serve ethically.

For the Bachelor of Science in Mechanical Engineering program, the admission requirements state that applicants must have an academic background of F.Sc Pre-Engineering or ICS with Mathematics and Physics from a recognized board or an equivalent qualification with at least sixty percent marks. Alternatively, a Diploma of Associate Engineer in a relevant field with a minimum of sixty percent marks is also accepted. The degree requirements mandate completing 136 total credit hours across 57 total courses, and students must maintain a minimum academic standing with a CGPA of 2.0 or higher. The Program Educational Objectives dictate that graduates are expected to exhibit specific attributes three to five years post-graduation. These include providing solutions to problems associated with design, manufacturing, and energy systems using the fundamentals of mechanical engineering, and designing and investigating mechanical systems by using analytical and modern techniques for innovation, technological development, and sustainability. Furthermore, graduates must acquire the competency to address real-world problems in socio-technological areas through ethical values and professional skills.

The undergraduate curriculum plan follows the Third Board of Studies Schema, where credit hour listings denote theory and lab hours. In the first semester, students take Applied Chemistry with no prerequisite for two and zero credit hours, Applied Physics with no prerequisite for two and one credit hours, Calculus and Analytical Geometry with no prerequisite for three and zero credit hours, Fundamentals of English Grammar with no prerequisite for two and zero credit hours, and Mechanical Engineering Drawing with no prerequisite for zero and two credit hours. The second semester consists of Workshop Practice Lab or Introduction to Computer Programming with no prerequisites for zero and two or two and one credit hours respectively, Basic Electronics with no prerequisite for two and one credit hours, and Engineering Materials and Metallurgy with no prerequisite for three and zero credit hours. This semester also requires Thermodynamics-I with no prerequisite for three and zero credit hours, Islamic Studies or Engineering Mechanics Statics with no prerequisites for two and zero or three and zero credit hours respectively, and Computer-Aided Grid Drawing Lab or Communication Skills with no prerequisites for zero and one or three and zero credit hours respectively. The third semester involves Differential Equations with a prerequisite of Calculus and Geometry for three and zero credit hours, Engineering Mechanics Dynamics and Lab with no prerequisite for three and zero and zero and one credit hours, Fluid Mechanics-I or Solid Mechanics-1 with no prerequisites for three and zero credit hours each, and Pakistan Studies or Electrical Engineering with no prerequisites for two and zero or two and one credit hours respectively.

The fourth semester covers Linear Algebra and Transforms with a prerequisite of Calculus and Geometry for three and zero credit hours, Fluid Mechanics II and Lab with a prerequisite of Fluid Mechanics-1 for three and zero and zero and one credit hours, and Machine Design-I or Engineering Economics with no prerequisites for three and zero or two and zero credit hours respectively. Additionally, students take Thermodynamics-II and Lab with a prerequisite of Thermodynamics-I for three and zero and zero and one credit hours, and Technical Report Writing and Presentation Skills with no prerequisite for two and zero credit hours. The fifth semester includes Solid Mechanics-II and Lab with a prerequisite of Solid Mechanics-1 for three and zero and zero and one credit hours, Computer-Aided Engineering Lab with no prerequisite for zero and one credit hours, Mechanical Instrumentation and Control with no prerequisite for two and one credit hours, Numerical Analysis with a prerequisite of Linear Algebra and Transforms for two and one credit hours, and Machine Design-II or Manufacturing Processes-I with no prerequisites for three and zero credit hours each. The sixth semester encompasses Manufacturing Processes-II and Lab with no prerequisite for three and zero and zero and one credit hours, Applied Statistics with no prerequisite for three and zero credit hours, Heat and Mass Transfer and Lab with no prerequisite for three and zero and zero and one credit hours, Internal Combustion Engines with no prerequisite for two and zero credit hours, and Mechanics of Machines and Lab with no prerequisite for three and zero and zero and one credit hours.

The seventh semester requires Maintenance Engineering with no prerequisite for two and zero credit hours, Refrigeration and Air Conditioning and Lab with no prerequisite for three and zero and zero and one credit hours, Health, Safety and Environment with no prerequisite for one and zero credit hours, Management Elective or Technical Elective-I with no prerequisites for three and zero credit hours each, and Project-I with no prerequisite for zero and three credit hours. The eighth and final semester concludes the undergraduate program with Mechanical Vibrations and Lab with no prerequisite for three and zero and zero and one credit hours, Professional Ethics and Entrepreneurship with no prerequisite for two and zero credit hours, Power Plant Engineering with no prerequisite for three and zero credit hours, and Technical Elective-II or Project-II with no prerequisites for three and zero or zero and three credit hours respectively.

For the Master of Science Program, the admission requirements include an academic background of sixteen years of education or an equivalent benchmark qualification, such as a four-year BS or B.E. degree, from an HEC-recognized university. Applicants must secure at least sixty percent marks under the annual grading setup or a minimum CGPA of 2.5 out of 4.0, though it is noted that the raw webpage text retains a copy-paste placeholder typo erringly naming Civil or Petroleum engineering fields within this admission block. Candidates must also provide a valid GAT General score with at least fifty percent marks. The degree requirements for the MS program consist of thirty total credit hours, which includes a total coursework requirement of eight advanced graduate courses, and students must maintain a minimum academic standing with a CGPA of 2.5 or higher.